Clawpressor

Manual. Free and open source by Studio 87. Support: ko-fi.com/studio87

Intro

Clawpressor is a four band compressor for bass, synth leads and vocals. Each band has its own compressor with a soft knee and a lookahead, plus a lift (upward compression) that brings up quiet detail, antialiased saturation and a stereo width control. A rhythmic gain pattern, a mix knob, a smooth soft clipper and a master limiter finish the chain.

It is free and open source, made by Studio 87. It collects nothing and works fully offline.

Quick start

- Put Clawpressor on a bass, a lead or a bus. Pick a preset from the list at the top.
- Drag the white crossover lines in the band display to set where the bands split.
- Pick a band and drag the white dot in its curve to set the threshold. Drag the yellow dot to set the lift floor.
- Use Mix to blend the processed sound with the original (parallel compression).
- Hover any control for one second to read what it does.

Bands

Four bands (low, low-mid, high-mid, high) split with phase-aligned Linkwitz-Riley crossovers. Each band has On, Mute and Solo. Threshold is the level above which the band is compressed, Ratio how strongly, Knee how gradually the compression starts. Attack and Release set the speed. Boost adds make-up gain after the compressor.

Lift, saturation and width

Lift floor and Lift ratio are an upward compressor: signals quieter than the floor are raised (a ratio of 2:1 lifts a signal 10 dB under the floor by 5 dB), fading out well below so silence is not boosted. Saturation adds antialiased tanh drive to the band and keeps the level. Width sets the stereo width of the band: 0 is mono (good for the low band), 100 is unchanged, 200 doubles the side.

Pattern and master

The pattern is 16 steps that dip the gain of the wet signal in time with the host tempo. Strength sets how deep, Speed how long a step is (1/16, 1/8, 1/4). Dry/Wet mixes processed and original. Lookahead lets the compressors see ahead of the audio (it adds the same latency to the dry path, reported to the host). The limiter release and the post gain finish the chain. Soft clip is a smooth knee clipper at the very end.

Recommendations

- Bass: put the low band on, set its Width to 0, add a little Saturation, ratio 4 to 6.
- Synth lead: ratio 3 to 4, a small amount of lift, widen the high bands to 120 to 160.
- Vocals: ratio 2 to 3 on the mid bands, slower attack, low band off.
- If the sound gets dull when you push it, lower the ratio before raising the threshold.
- Lookahead above 0 adds latency; turn it off when tracking live.

Not in this version

The three assignable pattern lanes from the design mockup (any parameter sequenced by 16 steps) are not in the plugin yet; the plugin has the single gain pattern.

Interface tour

Numbered pink badges mark every control. The list below gives the name and the full info-bubble text of each. The picture is taken from the design mockup that the editor follows.



#	Control	What it does
1	Previous preset	Previous preset.
2	Init	Open the preset list: factory presets and your own.
3	Next preset	Next preset.
4	Save	Save the current settings as your own preset (stays on this device).
5	■	Delete, export or import your presets.
6	Guide	Quick guide: what each view and control does.
7	Settings	Info bubbles and eco mode.
8	About & Support	About Studio 87, privacy, and ways to support the project.
9	Test signal	Pick what the preview rig plays into the plugin. 'Your file' uses the file you loaded.
10	Play	Start or stop the test signal. Shortcut: Space.
11	Load file	Load your own audio file (wav, mp3, flac, ogg). It stays in this browser tab and is never uploaded.
12	Input trim	Level into the plugin, like the gain of the track in your DAW. Factory presets expect the input to peak around -6 dB (see the marker on the meter). Double-click resets.
13	Compare: bypass	Hear the unprocessed signal at the same moment, to judge what the plugin adds. This is the preview rig, not a plugin control.
14	BPM 150	Tempo the plugin gets from the host. The pattern gain follows it.

#	Control	What it does
15	Bands	Spectrum before (grey line) and after (colour fill) with draggable crossover lines and the gain reduction of every band. Shortcut: 1.
16	Curve	Compressor curve of the selected band: input on the bottom, output up the side. Drag the corner to set the threshold and the end to set the ratio. Shortcut: 2.
17	Band scope	Waveform of the selected band: red is the original, green is after compression, grey is where they overlap. Shortcut: 3.
18	GR history	Gain reduction history of every band and the master limiter. Shortcut: 4.
19	Pattern	Pattern gain: draw a rhythmic volume pattern over 16 steps. Shortcut: 5.
20	Bands	How many bands the signal is split into. Drag the white crossover lines in the display to move them.
21	Crossover slope	Steepness of the crossover filters. Steeper keeps the bands cleaner. The plugin offers 6 to 96 dB/oct; 36 or 48 is a good start.
22	Whole ratios	Snaps every ratio to whole numbers (2:1, 3:1, 4:1 ...).
23	S	Solo: hear only this band (and any other soloed band).
24	M	Mute: silence this band.
25	Input	Level into this band's compressor. Double-click resets.
26	Thresh	Level above which the band is compressed. Also draggable on the Curve view. Double-click resets.
27	Ratio	How strongly the signal above the threshold is reduced. 'inf' makes it a limiter. Also draggable on the Curve view. Double-click resets.
28	Attack	How fast the compressor reacts. Short grabs transients, long lets punch through. Double-click resets.
29	Release	How fast the gain comes back. With Auto on, it follows the gain reduction: more reduction, slower release. Double-click resets.
30	Boost	Extra makeup gain after the compressor, on top of the automatic gain. Double-click resets.
31	Limit	Ceiling of this band's limiter (ratio infinity). Mockup limiter has no lookahead; the plugin will. Double-click resets.
32	Lim rel	Release of this band's limiter. Double-click resets.
33	Output	Level of this band after the limiter. Use it to rebalance the bands. Double-click resets.
34	Sat	Soft saturation of this band after the limiter (antialiased tanh). Drive and mix rise together. Warms bass and thickens leads. Double-click resets.
35	Width	Stereo width of this band (mid/side). 0 makes the band mono: set the low band to 0 for a mono bass. 100 is unchanged. Double-click resets.
36	Comp	Turns this band's compressor on or off. The band still passes through.
37	Auto rel	Release follows the gain reduction. The Release knob is ignored.
38	Soft knee	Soft knee eases into compression around the threshold. Off is a hard knee.
39	Limiter	Turns this band's limiter on or off.
40	Mix	Dry/wet of the whole plugin. 100 % is wet only. Around 75 % brings back dynamic peaks without losing much density (parallel compression). Default 50 %. Double-click resets.
41	Output	Level into the master limiter. Double-click resets.
42	Post gain	Level after the master limiter. Double-click resets.
43	Mstr rel	Release of the master output limiter. Double-click resets.
44	Master limiter	Safety limiter at the very end, after the mix, so peaks of the dry part are caught too. Ceiling -0.3 dB.
45	Autogain	Compressors add makeup gain automatically so loudness stays about the same when you compress harder.